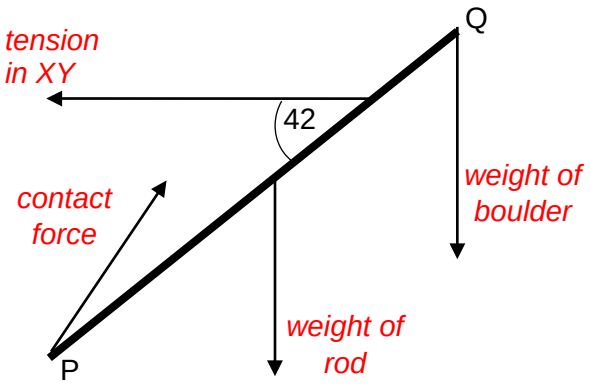
**2026 PU3 H2 Physics Prelim Paper 3 Suggested Answers**

| | | |
|---|---|---|
| **1(a)** | Impulse $= \Delta p$<br>$= p_{final} - p_{initial}$<br>$= -3.4 - 6.0$<br>$= \underline{-9.4 \text{ kg m s}^{-1}}$ (2 sf) | <br><br>**M1**<br>**A1** |
| **(b)** | F $= \Delta p / \Delta t$<br>$= -9.4 / 0.2$<br>$= -47$ N<br><br>acceleration $= -47 / 0.30$<br>$= -156.67$<br>$= \underline{-160 \text{ m s}^{-2}}$ (2 sf / accept positive answer here)<br>direction : opposite | <br><br>**C1**<br><br><br><br>**A1**<br>**A1** |
| **(c)** | Initial KE $= p_{initial}^2 / 2m$<br>$= 6.0^2 / (2 \times 0.30)$<br>$= 60$ J<br><br>Final KE $= p_{final}^2 / 2m$<br>$= 3.4^2 / (2 \times 0.30)$<br>$= 19.267$ J<br><br>Change in KE $= 60 - 19.267 = 40.733$ J $= \underline{41 \text{ J}}$ (0 dp) | <br><br>**C1**<br><br><br><br><br><br>**A1** |
| **(d)** | The collision is **inelastic** because the **kinetic energy of the ball decreases** after collision.<br><br>Can accept any of the following too:<br>• speed ball decreases<br>• momentum of ball decreases in magnitude<br>• relative velocities of separation is smaller than the relative velocities of approach<br>• energy of ball is not conserved | **B1** |

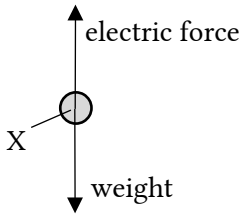
| | | |
|---|---|---|
| **2(a)** | **Resultant force** acting on the rod must be **zero** (*translational equilibrium*).<br><br>**Resultant torque / moment** of the rod **about any point** must be **zero** (*rotational equilibrium*). | **B1**<br><br>**B1** |
| **(b)** |  <br>1 mark for every 2 forces correctly drawn and labelled. | **A1**<br>**A1** |
| **(c)** | Using principle of moments, taking moments about P,<br><br>$(W_{boulder})(6.0 \cos 42^\circ)+(W_{rod})(6.0/2 \cos 42^\circ) = (T_{XY})\ (2.2)$<br>$17g\ (6.0 \cos 42^\circ) + 15g\ (6.0/2 \cos 42^\circ) = (T_{XY})\ (2.2)$<br>$T_{XY} = 487$ N<br>$= \underline{490\ \text{N}}$ (2 sf) | **C1**<br><br>**A1** |
| **(d)** | $T_{XY}$ will **increase**.<br><br>When the mass of the 'load' increases, the <u>sum of clockwise moments</u> due to weight of rod and weight of crate will <u>increase</u>, resulting in the need for <u>anticlockwise moments</u> due to $T_{XY}$ to <u>increase</u> as well. | **A1**<br><br>**M1** |

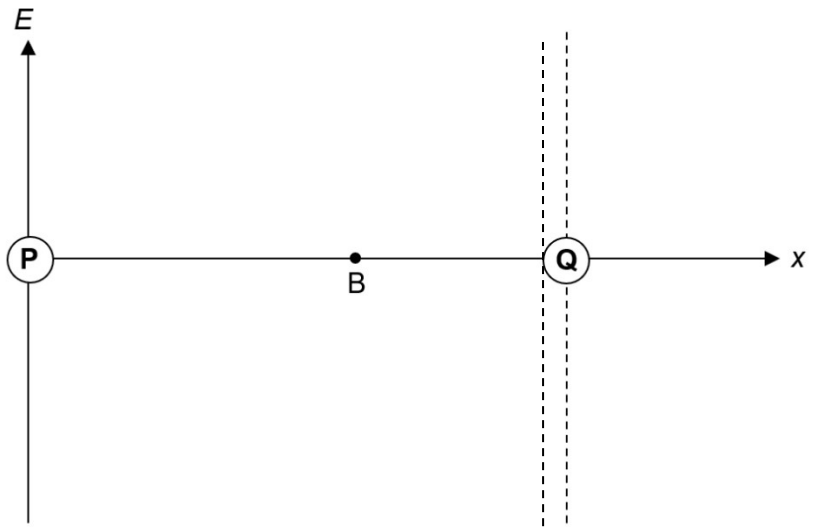
| | | |
|---|---|---|
| **3 (a)(i)** | As the space shuttle's **direction of velocity / motion is constantly changing**, the shuttle is accelerating.<br><br>For the shuttle to **accelerate**, a resultant force must be acting on it according to **Newton's second law of motion**. | **B1**<br><br>**B1** |
| **(ii)** | Gravitational force $F_G$ provides centripetal force $F_C$.<br><br>$F_G = F_C$<br>$GMm / (5R)^2 = mv^2 / 5R$<br>$v^2 = \frac{GM}{5R}$ | <br><br>**M1**<br>**A1** |
| **(iii)** | $E_K = ½ mv^2$<br>$= ½ m (\frac{GM}{5R})$<br>$= \frac{GMm}{10R}$ | <br>**M1** |
| **(b)** | Total Energy = KE + GPE<br>$= \frac{GMm}{10R} + (\frac{-GMm}{5R})$<br>$= \frac{-GMm}{10R}$<br>$= – G(7.8 \times 10^{24})(9.0 \times 10^4) /(10 \times 6.8 \times 10^6)$<br>$= – 6.8858 \times 10^{11}$ J<br>$=$ <u>$– 6.9 \times 10^{11}$ J</u> (2 sf) | <br><br>**C1**<br><br>**A1** |
| **(c)(i)** | At Z, $g_X = g_Y$<br>$GM/x^2 = G(2M) / (6.0 \times 10^8 – x)^2$<br>$(6.0 \times 10^8 – x)^2 / x^2 = 2$<br>$x = 2.4853 \times 10^8$ m<br>$=$ <u>$2.5 \times 10^8$ m</u> (2 sf) | <br>**C1**<br><br><br>**A1** |
| **(ii)** | At Z, GPE $= U_X + U_Y$<br>$= \frac{-GMm}{2.5 \times 10^8} + (\frac{-G(2M)m}{(6.0\text{-}2.5) \times 10^8})$<br>$= –4.5486 \times 10^{11}$ J<br>$=$ <u>$–4.5 \times 10^{11}$ J</u> (2 sf) | <br><br>**C1**<br><br>**A1** |
| **(iii)** | The space shuttle needs just enough energy to reach point Z, and the resultant gravitational force acting on it will be in the direction of planet Y.<br><br>extra KE needed = work done in moving from X to Z<br>$= [-4.5486 – (-6.8858)] \times 10^{11}$ J<br>$= 2.3372 \times 10^{11}$ J<br>$=$ <u>$2.3 \times 10^{11}$ J</u> (2 sf) | <br><br>**C1**<br><br>**A1** |

| | | |
|---|---|---|
| **4 (a)** | Upon release, the **elastic force** exerted by the spring **on the copper ball is larger** than the **copper ball's weight**.<br><br>Hence, the direction of the **resultant force is upwards**. | **B1**<br><br>**B1** |
| **(b)** | The additional elastic force experienced by the copper ball is due to the initial displacement of $x_0$. This force is also opposite in direction to the displacement $x$.<br><br>$\sum F = ma$<br>$ma = -kx$<br>$a = \frac{-k}{m}x$ | **C1**<br><br>**M1** |
| **(c)** | The copper ball's **acceleration is directly proportional to its displacement $x$ since $k$ is a constant** and<br>**the negative sign shows acceleration is always directed towards a fixed equilibrium position**. | **B1**<br><br>**B1** |
| **(d)(i)** | $\Delta\omega / \omega = ½ \Delta k/k + ½ \Delta m/m$<br>$= ½ (0.2 / 13.6) + ½ (0.005 / 1.200)$<br>$= \underline{0.0094}$ (2 sf) | **M1**<br><br>**A1** |
| **(ii)** | $\omega = \sqrt{\frac{13.6}{1.200}} = 3.3665$<br>$\Delta\omega = 0.03$ (1 sf)<br><br>Hence, $\omega = \underline{3.37 \pm 0.03}$ rad $s^{-1}$ | <br><br>**C1**<br><br>**A1** |
| **(iii)** | Total energy $= ½ m \omega_0^2 x_0^2$<br>$= ½ (1.200) (3.37)^2 (0.20)^2$<br>$= 0.27257$ J $= \underline{0.27}$ J | **C1**<br><br>**A1** |

| | | |
|---|---|---|
| **5 (a)(i)** | $V_X = \frac{R_X}{R_X + R_{TH}} E$<br><br>$P_X = V_X^2 / R_X$<br>$= (\frac{R_X}{R_X + R_{TH}} E)^2 / R_X$<br>$= \frac{R_X}{(R_X + R_{TH})^2} E^2$ | **C1**<br><br>**A1** |
| **(ii)** | When temperature decreases, **resistance of thermistor $R_{TH}$ increases**. This means that overall current in the **circuit will decrease**.<br><br>OR **Using the expression in (a)(i), since $R_{TH}$ increases,**<br><br>Hence, **$P_X$ decreases** since $P_X = I^2R_X$. | **M1**<br><br>**A1** |
| **(iii) (A)** | At balanced potential (when there is no current detected), it means that $E_S = V_{TH}$.<br><br>$E_S = V_{TH}$<br>$= \frac{R_{TH}}{R_X + R_{TH}} E$<br>$= \frac{2R_X}{R_X + 2R_X} E$<br>$= \frac{2}{3} E = 0.67E$ | **C1**<br><br>**A1** |
| **(iii) (B)** | When X is replaced with a resistor of lower resistance, it means that $V_{TH}$ will increase. **$V_{TH}$ will be greater than $E_S$**.<br><br>This will result in a **current flowing** in the secondary circuit, which will be **detected by the galvanometer**. | **B1**<br><br>**B1** |
| **(b)(i)** | $R = \rho l / A$<br>$= (4.9 \times 10^{-7})(0.800) / [(\pi/4)(0.027/100)^2]$<br>$= 6.8\ \Omega$ | **M1** |
| **(ii) (A)** | $R_{eff} = 3.4 + (1/1.7 + 1/6.8)^{-1}$<br>$= 4.76\ \Omega$<br><br>Current $= 6.0 / 4.76$<br>$= 1.26$<br>$=$ <u>1.3 A</u> (2 sf) | **C1**<br><br>**A1** |
| **(ii) (B)** | Voltmeter reading = <u>6.0 V</u> (equal to e.m.f. of circuit) | **A1** |

| | | |
|---|---|---|
| **6 (a)** | The alternating current causes a **change in the magnetic flux density** in the solenoid.<br><br>This causes a **change in the magnetic flux linkage in coil X**. According to **Faraday's law**, when there is a change in the magnetic flux linkage in coil X, an e.m.f. will be induced in the coil. | **B1**<br><br>**B1** |
| **(b)** | $B_{max} = u_0 nI$<br>$= (4\pi \times 10^{-7})(2500)(2.6)$<br>$= 8.1681 \times 10^{-3}$ T<br>$= \underline{8.2 \times 10^{-3}\ \text{T}}$ (2 sf) | <br><br><br>**A1** |
| **(c)** | $\Phi_{max} = NBA$<br>$= (100)(8.1681 \times 10^{-3})(7.4 \times 10^{-5})$<br>$= \underline{6.0 \times 10^{-5}\ \text{Wb-turns}}$ (2 sf) | <br>**C1**<br>**A1** |
| **(d)** | induced e.m.f. $= -d\Phi/dt$<br>$= -d(NBA)/dt$<br>$= -d(\Phi_{max} \sin 6\pi t)/dt$<br>$= 6\pi(\Phi_{max}) \cos 6\pi t$<br><br>Max e.m.f inducted $= 6\pi\ (6.0 \times 10^{-5})$<br>$= 1.13 \times 10^{-3}$ V<br>$= \underline{1.1 \times 10^{-3}\ \text{V}}$ (2 sf) | <br><br><br>**C1**<br><br><br>**A1** |

| | | |
|---|---|---|
| **7 (a)(i)** |  <br>1 mark for both forces correctly labelled with correct directions | **A1** |
| **(ii)** | Electric field strength is defined as the **force per unit positive charge**<br><br>**acting on a small stationary charge placed** at a point **in the electric field**. | **B1**<br><br>**B1** |
| **(iii)** | $F_E = F_G$<br>$qE = mg$<br><br>$E = mg / q$<br>$= (1.00 \times 10^{-6})\ (9.81) / (1.60 \times 10^{-12})$<br>$= 6.1 \times 10^{6}$ N $C^{-1}$ (2 sf) | **C1**<br><br>**M1** |
| **(iv)** | $E = \Delta V / d$<br>$\Delta V = E\ d$<br>$= (6.1 \times 10^{6})\ (5 \times 10^{-3})$<br>$= 30.500$ kV<br><br>$\Delta V = V_{bottom} - V_{top}$<br>$V_{bottom} = 30.500 + (-15.0)$<br>$=$ <u>15.5 kV</u> (2 sf) | **C1**<br><br>**A1** |
| **(v)(A)** | When separation between plates increases, the **electric field strength decreases**.<br><br>This results in a decrease in the electric force (acting upwards) on the particle. The **weight of the particle** (acting downwards) is now **greater than the electric force**, which will cause the particle's **path to be deflected downwards** / **towards the bottom plate**. | **B1**<br><br>**B1** |
| **(v)(B)** | There is **no change to the path** of the particle as the path is independent of the horizontal velocity of the particle. | **B1** |

| | | |
|---|---|---|
| **7(b)(i)** | The **electric field strength is equal** to the **negative potential gradient**. | **B1** |
| **(ii)** |  <br>1 mark for correct shape of the graph<br>1 mark for the cutting point B at $E$ = 0 | <br><br><br><br><br><br>**A1**<br>**A1** |
| **(iii)** | Work done $= q\,\Delta V$<br>$= (1.2 \times 10^{-9})\,(V_C - V_A)$<br>$= (1.2 \times 10^{-9})\,[-4 - (-10)]$<br>$= \underline{7.2 \times 10^{-9}}$ J | <br>**C1**<br><br>**A1** |
| **(iv)** | **Between A and B**, the potential gradient is positive, which means that the **resultant electric field strength is negative and pointing towards P**. **Between B and C**, the potential gradient is negative, which means that the **resultant electric field strength is positive and pointing towards Q**.<br><br>The only way for the above to happen is that **both charges on P and Q are negative**.<br><br>Alternative answer: can accept other reasonable answers that make use of data from Fig. 7.2 or 7.3. | **M1**<br><br><br><br>**A1** |
| **(v)** | From Fig. 7.2, $V_B$ = -1.90 V.<br><br>$V_B = V_{\text{due to P}} + V_{\text{due to Q}}$<br>$-1.90 = \dfrac{1.5Q}{4\pi\varepsilon_0(0.0275)} + \dfrac{Q}{4\pi\varepsilon_0(0.0225)}$<br><br>Solving for Q,<br>Charge on sphere Q = $\underline{2.13 \times 10^{-12}}$ C<br>Charge on sphere P = $\underline{3.20 \times 10^{-12}}$ C | <br><br>**C1**<br>**C1**<br><br><br>**A1** |

| | | |
|---|---|---|
| **8(a)(i)** | Polarisation is where the **oscillations in a wave are confined to one direction only in a plane** normal to the direction of transfer of energy of the wave. | **B1** |
| **(ii)** | **The student is wrong**.<br><br>Sound waves are **longitudinal** in nature. The **particles** in the medium (usually air) **vibrate parallel to the direction of wave propagation** and hence cannot be polarised. | **A1**<br><br>**M1** |
| **(iii)** | Intensity = $\underline{2I_0}$ | **A1** |
| **(iv)** | Using Malu's Law,<br>$I_1 = I_0 \cos^2(30^\circ - 15^\circ)$<br><br>$I_2 = I_1 \cos^2(50^\circ - 30^\circ)$<br>$= I_0 \cos^2(15^\circ) \cos^2(20^\circ)$<br>$= 0.82387 I_0$<br>$= \underline{0.82 I_0}$ (2 sf) | **C1**<br><br>**A1** |
| **(b)(i)** | distance between slits, $d$ = 1 / (4.5 x $10^5$)<br>= 2.222 x $10^{-6}$ m<br><br>tan $\theta$ = 0.60 / 2.0<br>$\theta$ = 16.699°<br><br>For diffraction grating,<br>$d \sin\theta = n\lambda$<br>$\lambda$ = 2.222 x $10^{-6}$ sin 16.699°<br>= 640 nm | **C1**<br><br>**M1** |
| **(ii)** | Assume $\theta$ = 90° to find maximum number of bright spots.<br><br>(2.222 x $10^{-6}$) sin 90° = $n$ (640 x $10^{-9}$)<br>$n$ = 3.47<br><br>This means there are 3 bright spots above the central spot, and 3 bright spots below the central spot.<br><br>In all, there will be 3 + 3 + 1 = $\underline{\text{7 bright spots}}$. | **C1**<br>**A1**<br><br>**A1** |

| | | |
|---|---|---|
| **(iii)** | A **bright white fringe is observed at the centre** of the screen, because **all wavelengths of light reach the screen with zero path difference**.<br><br>The **other bright fringes consist of a continuous spectrum of colours**, because **different wavelengths of light diffracted at different angles**.<br><br>For every spectrum of colours, **violet colour is always closer to the centre and red is the furthest away**. This is because **shorter wavelengths diffract at smaller angles** OR **longer wavelengths diffract at larger angles**<br><br>*1 mark can be awarded if 2 of the observations are correctly made but the explanations are inadequate.* | **B1**<br><br>**B1**<br><br>**B1** |
| **(c)(i)** | From the graph, the distance between two successive bright fringes is **5.0 x $10^{-3}$ m**.<br><br>For double slit experiment,<br>$a = \lambda D/x$<br>$= (640 \times 10^{-9})(2.0) / (5 \times 10^{-3})$<br>$= \underline{2.56 \times 10^{-4}\ m}$ (2 sf) | **C1**<br><br>**A1** |
| **(ii)** | From the graph, X shows the second constructive interference. This means that the path difference of the two waves at X is 2λ.<br><br>Hence, path difference $= 2.0 \times 640 \times 10^{-9}$<br>$= \underline{1.3 \times 10^{-6}\ m}$ (2 sf) | **A1** |
| **(iii)(A)** | Destructive interference takes place at Y. Hence when one of the slits is partially blocked by the tracing paper, the **light wave's intensity / amplitude will be decreased**. As a result, at Y, **there will not be complete destructive interference**.<br><br>**The dark fringe at Y will now be brighter**. | **B1**<br><br>**A1** |
| **(iii)(B)** | **There will not be any change** (distance between fringes is not affected by amplitude / intensity of light coming through the slits). | **A1** |